Architecture: representation & segmentation stack

The canonical, frozen shape (2026-06-23). A falsifiable multi-view pipeline for bioacoustic structure

See also: [`BENCHMARK.md`](BENCHMARK.md) (the numbers), [`CLAUDE.md`](CLAUDE.md) (invariants), [`configs/

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1. The stack (data flow)

audio / annotations

↓ segm

Two representation layers, deliberately distinct:

layer	class	role
event carrier	`DualStreamSegment` (`representation/dua	the typed-event **spine** — one segment
analysis unit	`DualStreamVocalisation` (`representat	the **call-level unit** diagnostics + ab

Both are built from the same `CallEvent`; the carrier is the lower-level typed-event sequence

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2. Segmentation router (per regime)

`configs/segmentation_router.yaml` + `run_pipeline(segmentation=...)` in `orchestration.py`.

regime	method	HECB role	F1
discrete calls in noise	PCEN+CRNN (`segmentation/hecb.py`)	hecb_annotator	meerkat 0.70, marmoset 0.88
continuous speech-like	NBD+CPC calibrator (`segmentation/nbd_*.	optional	LibriSpeech 0.836 combined
exploratory / unknown	NBD + ABSTAIN	none	conservative

`segmentation/hecb.py:hecb_segment_audio()` → `CallEvent`s; `hierarchical.py:hecb_decode()` (snap=Fa

3. Segmentation objects

`CallEvent` (`hierarchical.py`) — one detected vocalization: onset/offset, CRNN confiden

4. Representation objects

`DualStreamSubToken` (`vocalisation.py`) — typed internal child: relative_start/end (normali

Builders: `build_vocalisation()` (vocalisation) and `build_subcall_representations()` (subcall) c

5. Guardrails (do not weaken)

1. **Ethological firewall** (`embeddings/firewall.py` + `configs/model_metadata.json`) — cross-sp

6. Canonical claim (paper-safe)

Each vocalisation is represented as a **hierarchical dual-stream event**: a whole-call parent tol

7. Key entry points

Segment (any regime) → CallEvents with internal segments

from bioacoustic_pipeline.segmentation.hecb import hecb_segment_audio, load_hecb

from bioa

CallEvents → canonical analysis unit

from bioacoustic_pipeline.representation.vocalisation import (

build_v

DUSTED recurrent subsequence motif discovery (cross-species motif arc)

from bioacoustic_pipeline.diagnostics.motif_miner.subsequence_motifs import (

CallUni

Or the full pipeline

from bioacoustic_pipeline.orchestration import run_pipeline

report, tra

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8. DUSTED recurrent subsequence motif discovery (`motif_miner/subsequence_motifs.py`)

Adapts spoken-term-discovery (DUSTED) to **recurrent acoustic terms** in animal

vocalisati

Design

Each vocalisation = one `CallUnits` (ordered internal sub-call units). Windows of

length L=

method	matching primitive	role
`exact_ngram_clusters`	exact cluster-ID tuple match	fragments every variant; highest precisi
`fixed_length_cosine_clusters`	same-length windows, mean per-position c	catches substitutions, misses indels
`discover_recurrent_subsequences`	**DTW with indels** (role-aware, length-	catches substitutions + indels; richest

DTW cost (biologically grounded in the onset/body/offset positional-role result):

substitution(i, j) = (1 − cosine(fp_i, fp_j)) + role_penalty·𝟙[role_i ≠ role_j]

Constrained cascade (full all-pairs DTW is never run): window generation →

centroid-c

Diagnostics (per motif family + corpus summary)

`label_purity` — modal cluster-tuple fraction among members (GT validation when

clusters

Terminology (relanguage)

Speech word discovery is adapted, but the animal target is **recurrent acoustic

structure*

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9. Cross-corpus motif arc (5 animal corpora + LibriSpeech)

The DUSTED module runs across 5 animal corpora with identical diagnostics, producing a

cross-spe

corpus	regime	inventory	key finding	§ref
Bengalese finch	discrete syllable song	diverse, high-purity	fixed-cosine wmean purity 0.895; DTW rec	§12
Egyptian fruit bat	graded bout vocalisations	graded/intermediate	Emitter V ~0.46 ≫ Context V ~0.16 (2.8×)	§13
InfantMarmosetsVox	discrete calls (oracle HEGP)	sparse, one dominant	DTW calltype V 0.51 = 4.1× exact; degene	§11
RookID	noisy outdoor corvid calls	degenerate (one dominant)	Event V 0.22 > Source V 0.18; motifs are	§18–§19
LibriSpeech	human speech (controlled)	diverse word structure	fingerprint leads consistently; clusteri	§14–§16

The difference is the corpus/repertoire, not the method — DUSTED behaves correctly in

all regime

Individual identity across species

The 44-D acoustic fingerprint consistently preserves individual identity:

species	identity target	fingerprint V / accuracy	held-out control
Egyptian fruit bat	Emitter	Emitter V ~0.46 (motif families)	70/30 held-out: V 0.375 (§17)
RookID rook	Source (individual)	macro-F1 0.287 (4× chance), V 0.384	recording-date held-out (§18)
InfantMarmosetsVox	Caller	caller V present, sub-call augmentation	twin-pair held-out

Sub-call structure does NOT improve individual-ID accuracy (RookID: whole-call wins

on all met

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10. H0-CMC framework + speaker-invariant representation

science (2026-06-23)

H0-CMC: Causal Multimodal Convergence

The pipeline's Track C validation framework, refined from H0-Unified. See

[`h0-cmc-
[`cpc-poli

One-line: *Animal vocalisations are structured acoustic cues whose functional

interpreta

4-stage evidence ladder: L0 (acoustic association) → L1 (contextual prediction) →

L2 (receiv

Explicit-vs-implicit model test: plugged directly into the bat §9 A–G ablation —

D (acoust

Speaker-invariant representation science (human-speech side)

Cross-speaker same-word retrieval (§25) + DBA consensus-template sweep (§27) prove:

Word str

Cross-species speaker↔caller analogy (§25 ↔ §26)

Human **words** generalize across speakers (5–7× chance); animal **call-types** barely

generalize

CPC selective-use policy

Animal pipeline = **fingerprint + HECB + DUSTED + Track C** (no species CPC).

CPC = **h

Architecture v1.0 — Complete (2026-06-23)

The full 12-step roadmap is implemented, tested (247 tests), and committed. Here is the

complete

Core pipeline (unchanged)

`orchest

Segmentation router (the centrepiece — v1.0 frozen)

`confi
regime_cl

Structural discovery (multi-paradigm, v1.0 frozen)

`diagnos

H0-CMC evidence ladder

`diagn
classify_ti

Track C playback

`experin

Hardening

`verdict_

The three validated invariants

1. **Route

The three-bucket thesis (the paper's mature claim)

Every finc